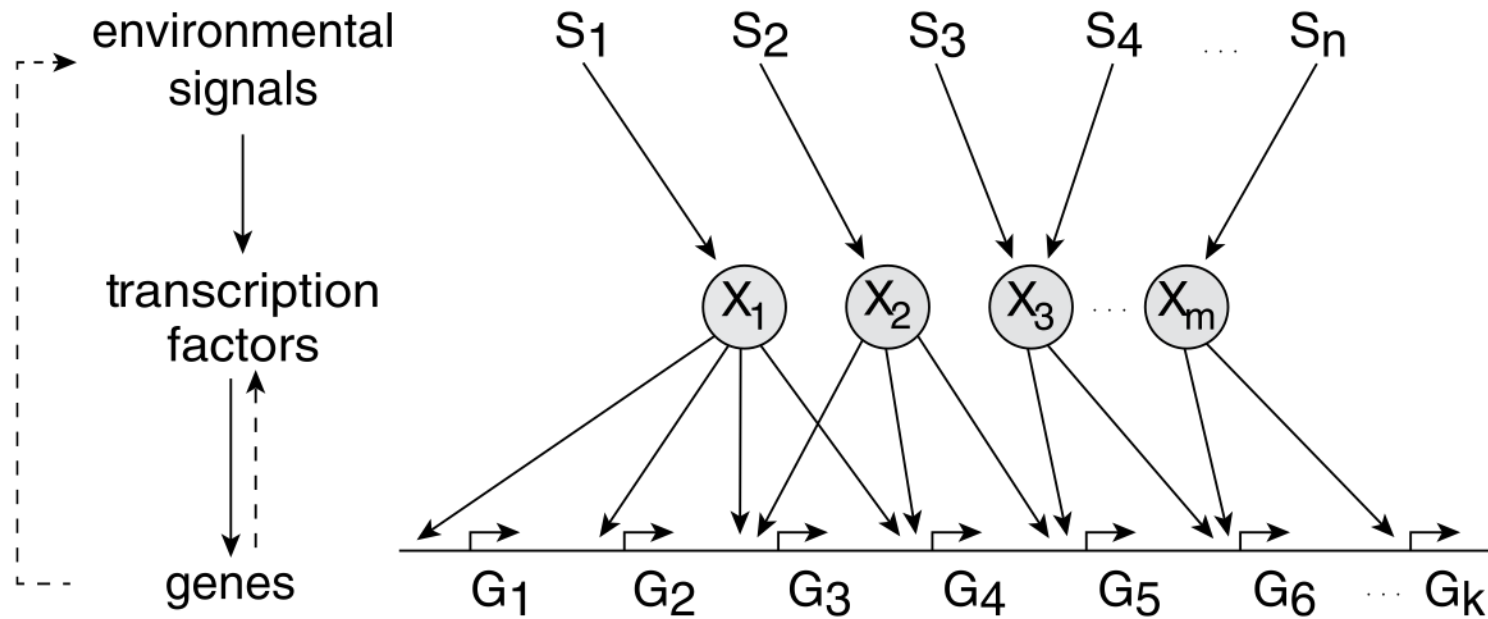
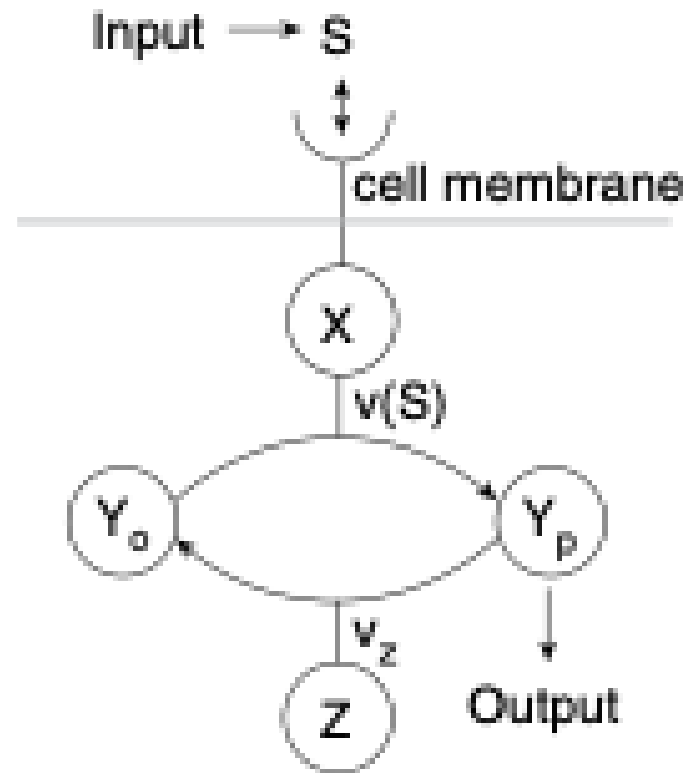


Lecture 1 Chemotaxis

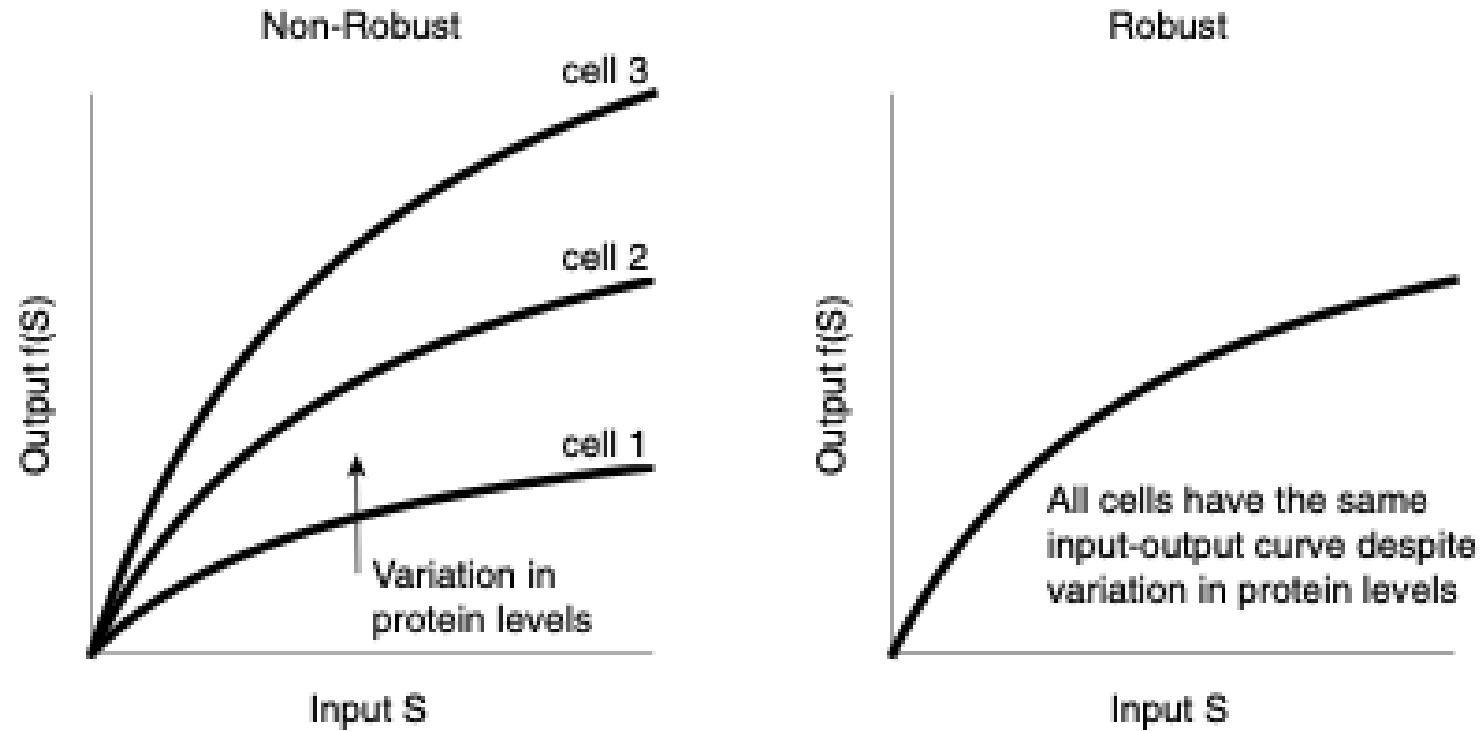
How do cells think ?



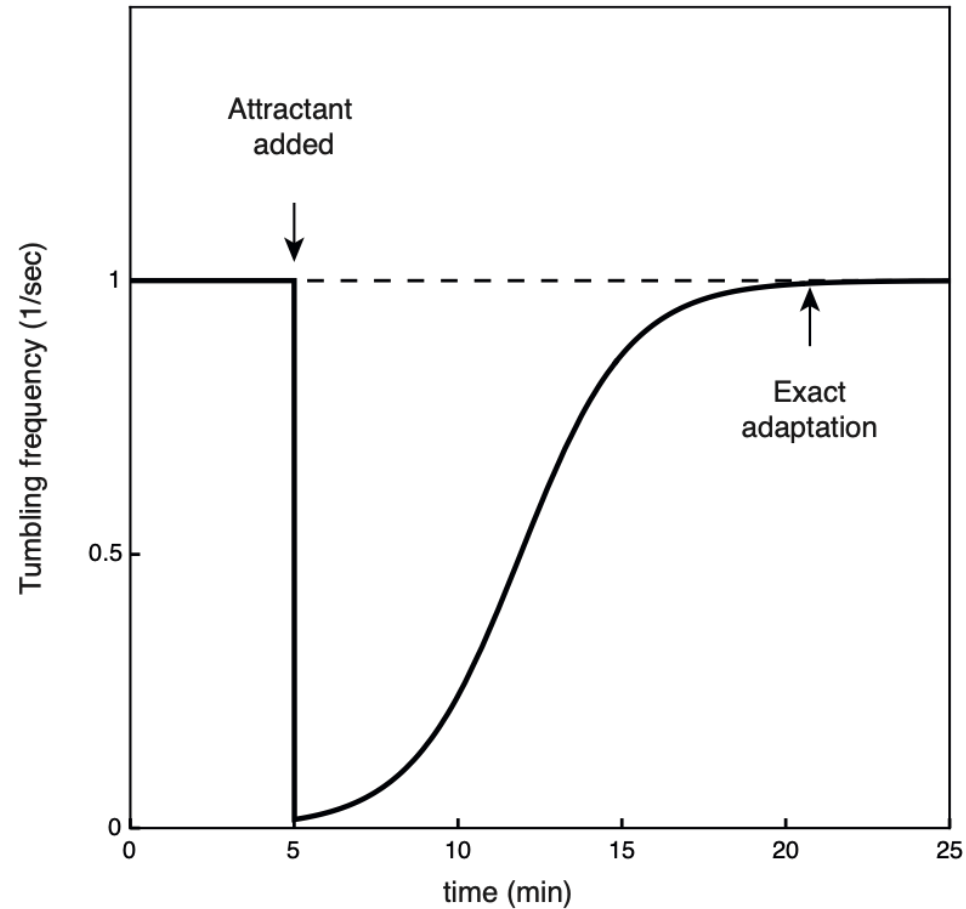
Simple models of signal-transduction circuits are not robust



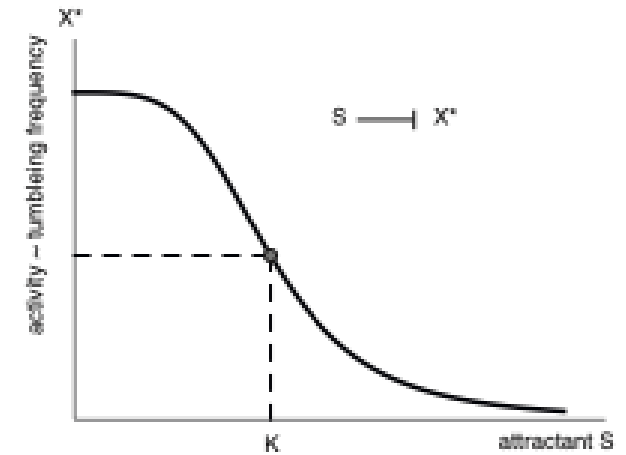
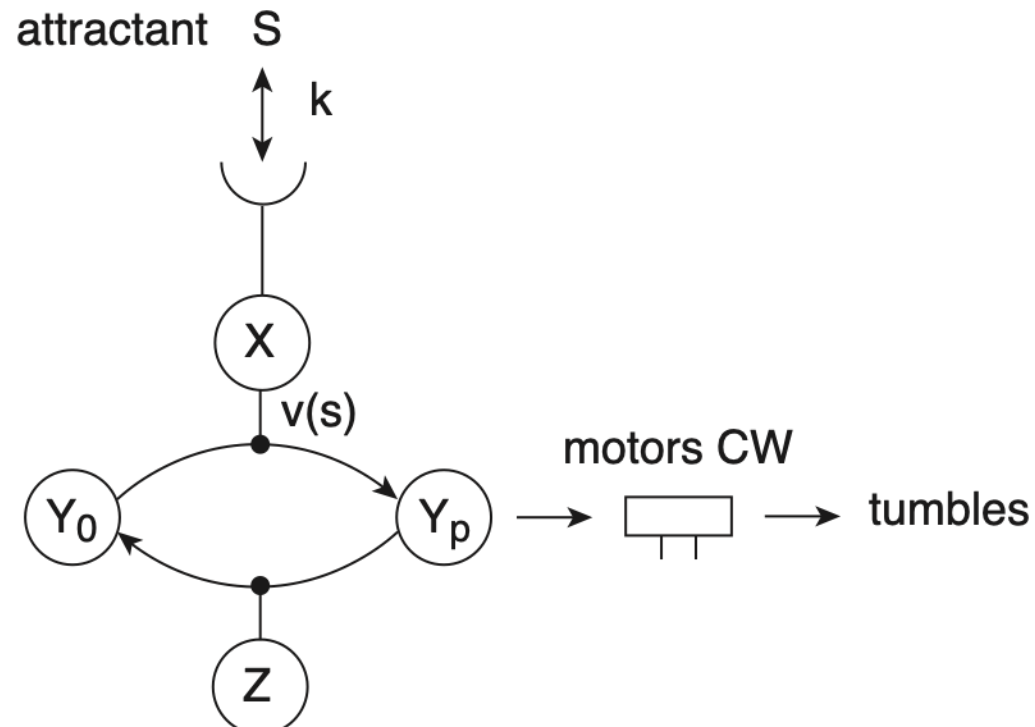
Biological circuits must operate robustly in the face of variations



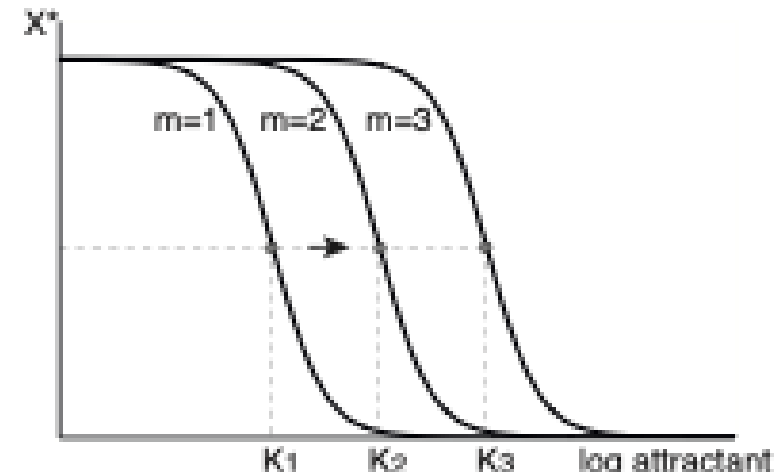
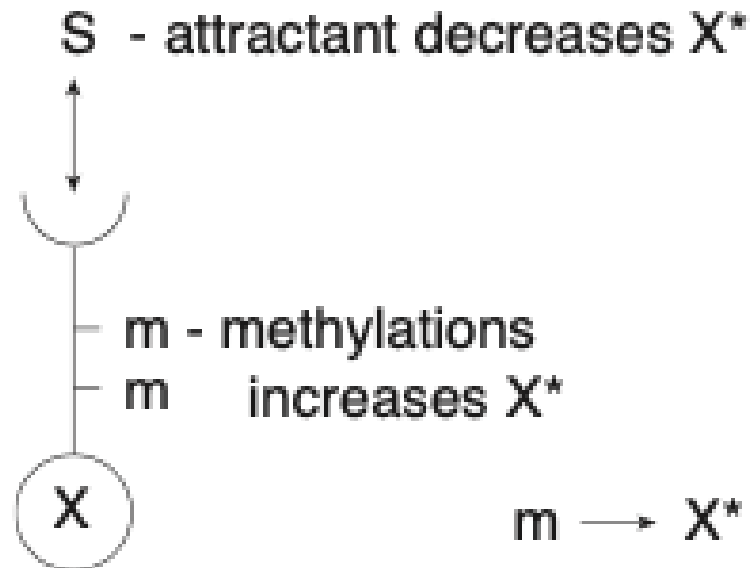
A basic feature of the chemotaxis response can be revealed by a simple experiment



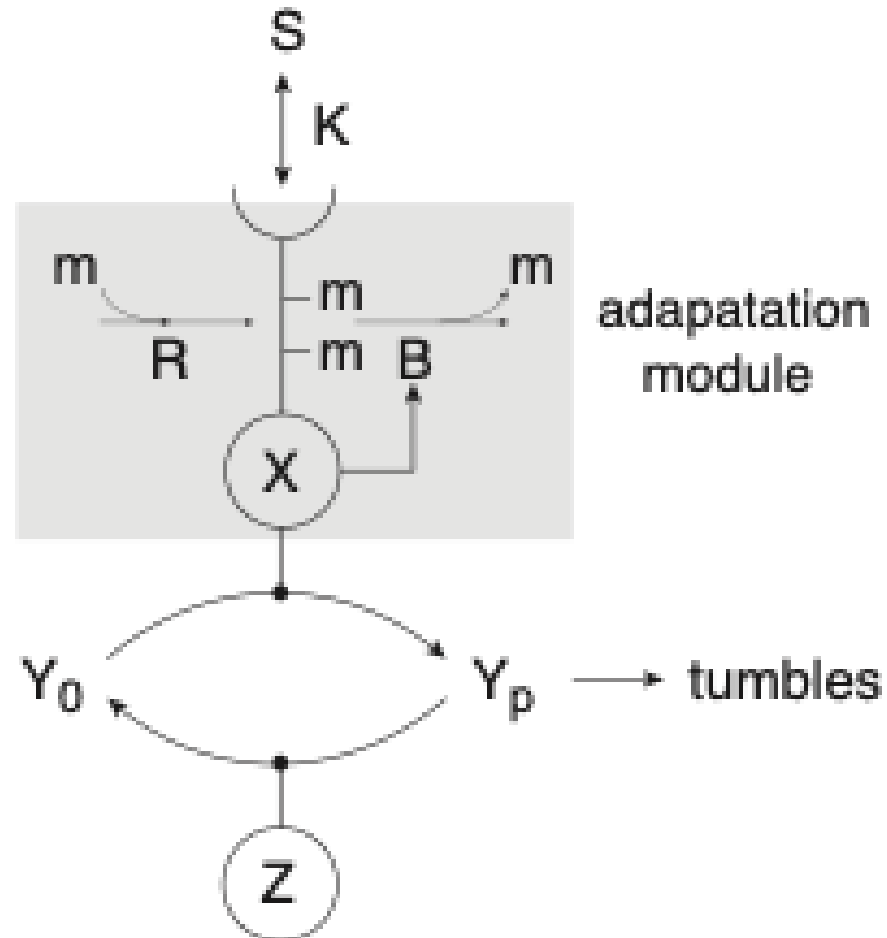
Attractant lowers the activity of X



Adaptation is due to slow modification of X that increases its activity

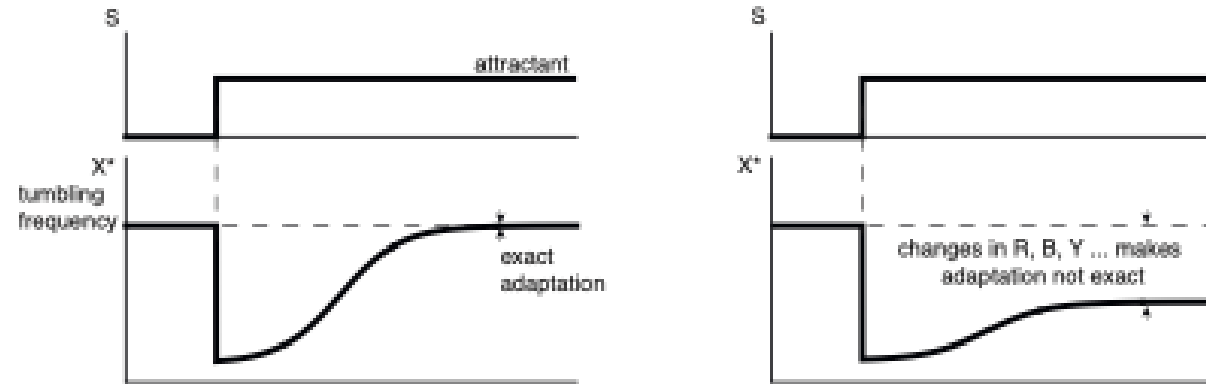


CheR and CheB form an adaptation module

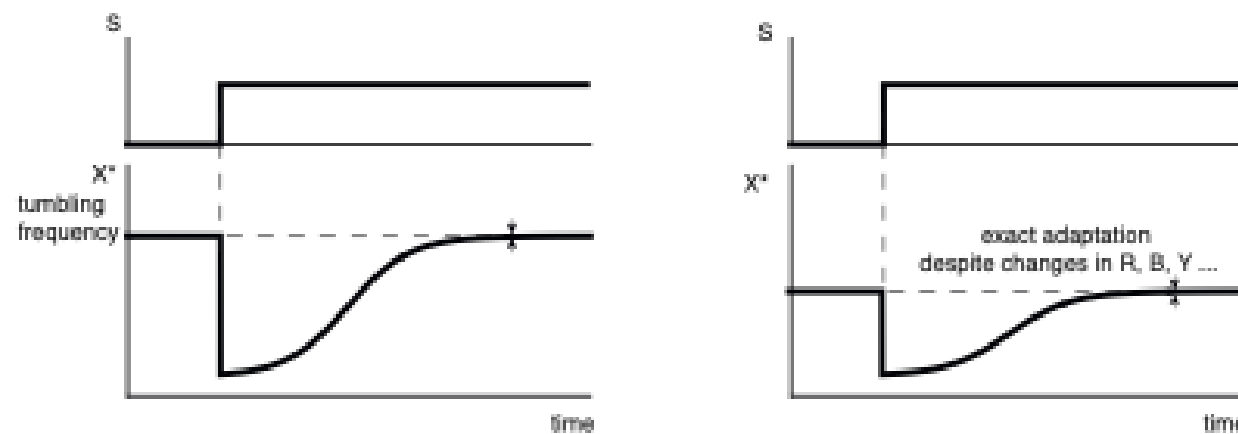


Many model adapt – but only some do so robustly

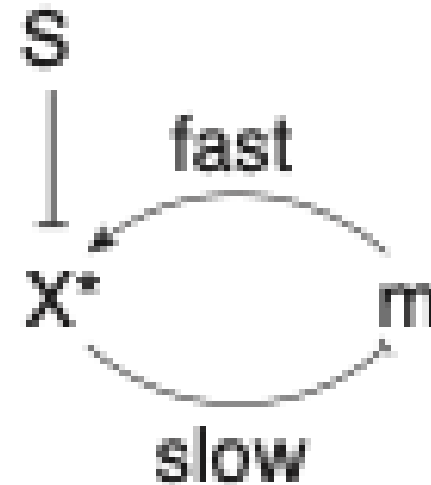
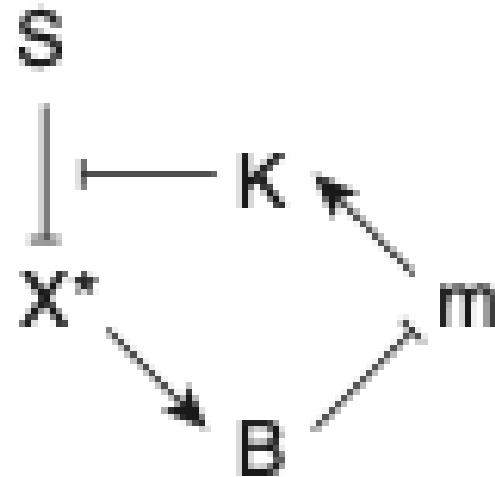
Non robust model



Robust model



Barkai–Leibler's key proposal: B demethylates only active X, forming a slow negative feedback loop

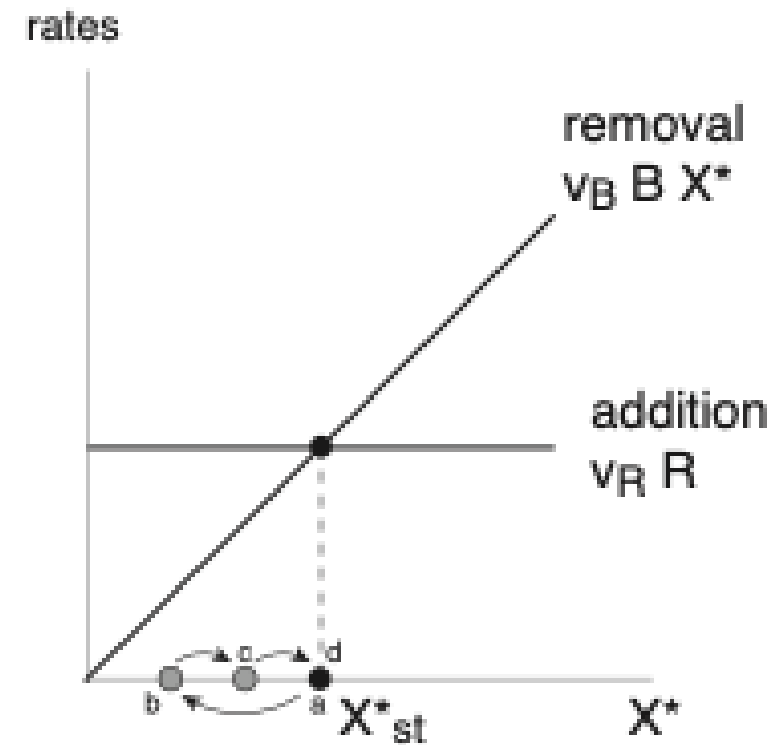
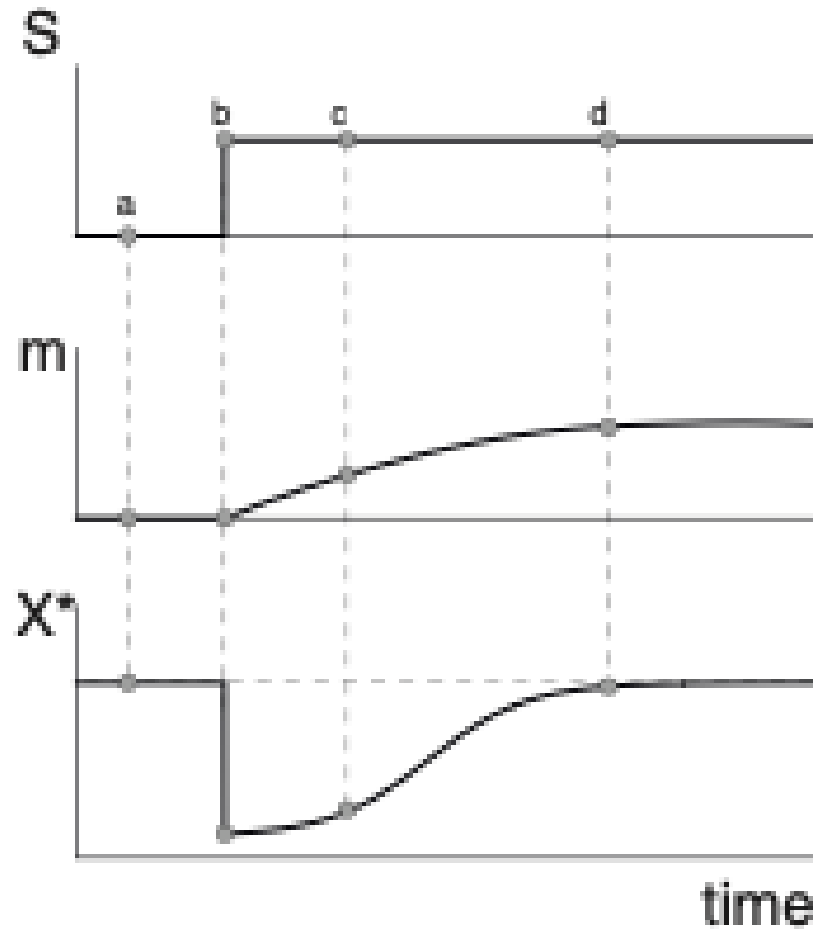


The Barkai-Leibler model explains robust exact adaptation

$$dm/dt = V_r R - V_b B X^*$$

$$X^*_{st} = V_r R / V_b B.$$

Exact adaptation is robust; baseline activity and adaptation time are not



Barkai-Leibler model is mapped to integral feedback

$$\frac{dm}{dt} = VbB(X_{st}^* - X^*)$$

$$m(t) \sim \int_{-\infty}^t (X_{st}^* - X^*(t)) dt \sim \int error(t) dt$$

Alon et al. experimentally test robust exact adaptation

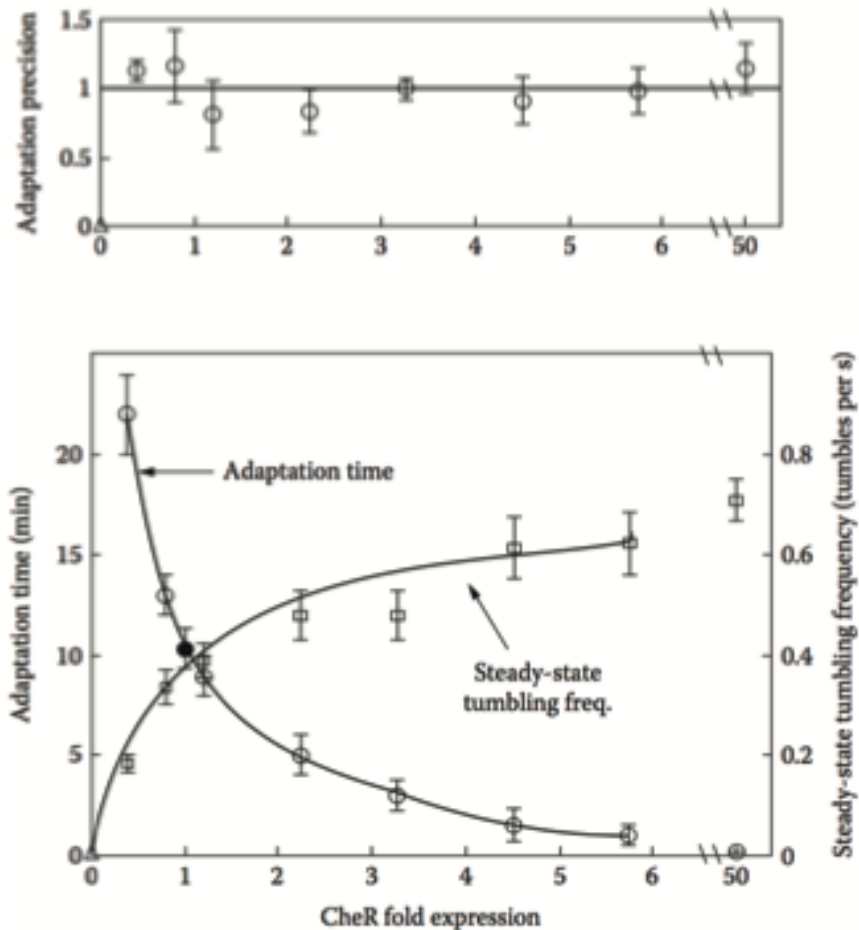
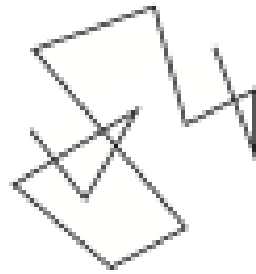


Table 1 Effect of variation in chemotaxis protein levels on behaviour

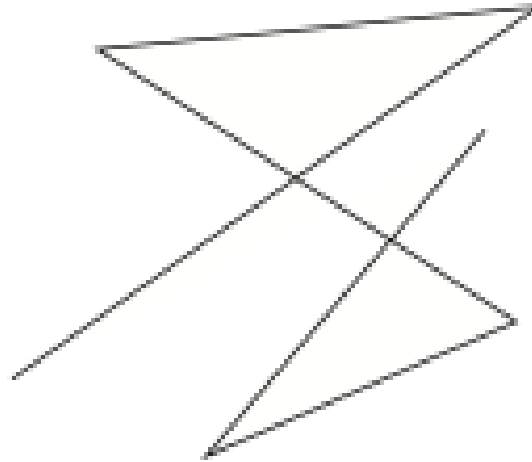
Protein varied	Fold expression	Strain background	Steady-state tumbling frequency (s^{-1})	Adaptation time (min)	Precision of adaptation
Wild type	1.0	Wild type (RP437)	0.44 ± 0.03	10 ± 1	0.98 ± 0.05
CheB*	0.4 ± 0.1	$\Delta cheB$ (RP4872)	0.66 ± 0.06	7 ± 1	0.98 ± 0.12
CheB*	12 ± 3	$\Delta cheB$ (RP4872)	0.34 ± 0.02	15 ± 1	1.09 ± 0.11
CheB ^{0.1}	~ 11	$\Delta cheB$ (RP4872)	0.74 ± 0.06	9 ± 2	0.90 ± 0.13
CheY*	0.2 ± 0.1	$\Delta cheY, Z$ (RP6231)	0.24 ± 0.04	11 ± 3	1.04 ± 0.08
CheZ	0	$\Delta cheZ$ (RP1616)	1.6 ± 0.1	10 ± 2	1.1 ± 0.34
Tar, Tap, CheR, B, Y, Z Φ	5 ± 2	Wild-type (RP437)	0.30 ± 0.06	3 ± 1	1.04 ± 0.07

Source: Alon, An Introduction to Systems Biology, 2nd ed. (2020)
U. Alon et al 1999 Nature

Protein concentration fluctuations result in individuality in chemotaxis



rapid progress
in dense obstacles



rapid progress
in free liquid